



Modelling Financial Markets

Seminar 9: Realized Volatility — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BV: Bipower Variation
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ DM: Diebold–Mariano (test)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-CJ: HAR model with Continuous and Jump components
- ▣ JB: Jarque–Bera (test)
- ▣ MAE: Mean Absolute Error
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ NW: Newey–West (standard errors)
- ▣ OLS: Ordinary Least Squares
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ RV: Realised Variance
- ▣ SE: Standard Error
- ▣ TQ: Tripower Quarticity
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility IndeX

Outline

Introduction

Part A: Pen and Paper

Part B: Computation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Compute realised and bipower variation step by step and test a day for jumps
 - ▣ Quantify the bias from microstructure noise and choose a sampling frequency
 - ▣ Estimate HAR models with robust standard errors and forecast variance
 - ▣ Compare variance forecasts with QLIKE, Diebold–Mariano and Mincer–Zarnowitz
-
- ▣ Background: Lecture 9; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight pen-and-paper exercises (A1–A8)
- ▣ **Part B:** eight computational exercises with inference on SPY and Bitcoin (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Units: returns in %, daily variances in %²; annualise with 252 days (SPY) or 365 days (Bitcoin)
- Every daily realised measure is itself an estimate: report how precise it is

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (RV and BV step by step), A3 (noise bias and optimal sampling), A5 (a log-HAR forecast), A6 (QLIKE and MSE)
- Part B: B1 (jump test on SPY), B3 (sampling frequency with block bootstrap), B5 (log-HAR with Newey–West errors)

Proposed: you solve them

- Part A: A2 (a jump in six returns; model: A1), A4 (two-scale RV; model: A3), A7 (two-day forecast; model: A5), A8 (DM and MZ step by step; model: A6)
- Part B: B2 (Bitcoin jumps; model: B1), B4 (standardised returns; model: B3), B6 (5-day forecasts; model: B5, A6, A8), B7 (HAR-CJ; model: B5, B1), B8 (roughness; model: B3)
- Part C: C1, open question and project seed; C2, critique of an AI answer

Setup: Python Environment and Data

-  **Open in Colab:** `chapter9_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `arch`, `matplotlib`
- SPY, 5-minute bars of the regular US session, October 2020 – September 2026; Bitcoin, 5-minute bars, 24-hour UTC days, September 2024 – September 2026
- Daily closes of SPY (adjusted) and Bitcoin for GARCH; SPY total daily variance = intraday RV + squared overnight return

Formulas Used Today

- $RV_t = \sum_{i=1}^M r_{t,i}^2$; $BV_t = \frac{\pi}{2} \frac{M}{M-1} \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}|$; $se(RV_t) = \sqrt{\frac{2}{3} \sum_i r_{t,i}^4}$
- Jump test [Huang & Tauchen \(2005\)](#): $z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\theta M - 1 \max(\mathbf{1}, \mathbf{TQ}_t/BV_t^2)}}$, $\theta = \pi^2/4 + \pi - 5$
- Noise: $\mathbb{E}[RV^{(n)}] = IV + 2n\omega^2$; $n^* = (IQ/(4\omega^4))^{1/3}$ [Bandi & Russell \(2008\)](#); two-scale $RV = RV^{\text{avg}} - \frac{\bar{n}}{n} RV^{\text{all}}$ [Zhang, Mykland & Aït-Sahalia \(2005\)](#)
- log-HAR [Corsi \(2009\)](#): $\ln RV_{t+1} = \beta_0 + \beta_d \ln RV_t + \beta_w \overline{\ln RV_t^{(5)}} + \beta_m \overline{\ln RV_t^{(22)}} + \varepsilon_{t+1}$
- QLIKE = $RV/F - \ln(RV/F) - 1$ [Patton \(2011\)](#); DM: $t = \bar{d}/se_{\text{NW}}(\bar{d})$ [Diebold & Mariano \(1995\)](#); MZ: $RV_t = a + bF_t + u_t$, $H_0 : a = 0, b = 1$

A1: Realised and Bipower Variation, Step by Step [Solved]

Task

- ▣ Six consecutive 5-minute returns (%):
0.08, -0.15, 0.03, 0.10, -0.04, 0.12
- ▣ (a) Realised variance of the half hour
- ▣ (b) Bipower variation and the share $(RV - BV)/RV$
- ▣ (c) RV from two 15-minute returns; annualised volatility if the whole day ($M = 78$) looked like this half hour

Solution

- ▣ Squares: 0.0064, 0.0225, 0.0009, 0.0100, 0.0016, 0.0144;
 $RV = 0.0558$
- ▣ $|r_i||r_{i-1}|$: 0.0120, 0.0045, 0.0030, 0.0040, 0.0048; sum
0.0283
- ▣ $BV = \frac{\pi}{2} \cdot \frac{6}{5} \cdot 0.0283 = 0.0533$; $(RV - BV)/RV = 0.044$:
no sign of a jump
- ▣ 15 minutes:
 $(0.08 - 0.15 + 0.03)^2 + (0.10 - 0.04 + 0.12)^2 = 0.0340$:
one noisy number from two returns
- ▣ $\sqrt{252 \times 0.0558 \times 78/6} = 13.5\%$ p.a.

A2: A Jump in Six Returns [Proposed]

Solution discussed in class

Task

- Same returns as A1, but the last one is -0.90% (a news jump)
- (a) RV, BV and the jump part $RV - BV$
- (b) The ratio statistic z with $M = 6$, given $TQ = 0.0097$; reject at the 0.1% level?
- (c) Why can the test miss a jump that is most of the variation?
- Model: A1

A3: Noise Bias and Optimal Sampling [Solved]

Task

- Daily IV = $1\%^2$; i.i.d. noise with $\omega = 0.01\%$; session of 23 400 seconds
- (a) $\mathbb{E}[RV]$ with returns every 5 minutes, 1 minute, 1 second
- (b) MSE-optimal n^* with $IQ = IV^2$, and the implied interval
- (c) Which interval would you use in practice, and why?

Solution

- $n = 78$: $1 + 2 \cdot 78 \cdot 0.0001 = 1.0156$; $n = 390$: 1.0780;
 $n = 23\,400$: 5.6800
- $n^* = (1/(4 \cdot 10^{-8}))^{1/3} = 292$: one return every
 $23\,400/292 = 80$ seconds
- 1–5 minutes: near n^* , bias below 8%; the MSE is flat
around n^* and 5 minutes is robust to a larger ω

A4: A Less Liquid Stock [Proposed]

Solution discussed in class

Task

- A BVB stock with $IV = 1\%^2$ and noise $\omega = 0.03\%$ (wider spread, larger tick)
- (a) Bias at 5 minutes and at 1 minute; optimal n^* and sampling interval
- (b) One day: $RV^{\text{all}} = 1.70$ from 1-minute returns ($n = 390$), average over 5 shifted 5-minute grids $RV^{\text{avg}} = 1.10$; compute two-scale RV, with and without the small-sample factor
- (c) Estimate ω^2 from RV^{all}
- Model: A3

A5: A log-HAR Forecast [Solved]

Task

- ▣ Rounded SPY log-HAR: $\hat{\beta} = (-0.08, 0.36, 0.34, 0.17)$
- ▣ Today: $RV_t = 2.0$; mean of log RV over the week $\ln 1.2$ and over the month $\ln 0.9$
- ▣ (a) Tomorrow's forecast, in $\%^2$ and annualised
- ▣ (b) Weights on lag 1, on each of lags 2–5, on each of lags 6–22
- ▣ (c) Persistence and the long-run level implied by the model

Solution

- ▣ $-0.08 + 0.36(0.693) + 0.34(0.182) + 0.17(-0.105) = 0.214$;
 $e^{0.214} = 1.238\%^2$, i.e. 17.7% p.a.
- ▣ $w_1 = 0.36 + 0.34/5 + 0.17/22 = 0.436$; $w_{2..5} = 0.076$;
 $w_{6..22} = 0.008$
- ▣ Persistence $0.36 + 0.34 + 0.17 = 0.87$
- ▣ Long-run level $\exp(-0.08/(1 - 0.87)) = 0.540\%^2$ (11.7% p.a.; before the variance correction)

A6: QLIKE and MSE Step by Step [Solved]

Task

- Realised variance on three days: 0.8, 1.5, 0.6
- Forecast A: 1.0, 1.0, 1.0 (constant); forecast B: 0.6, 1.6, 0.4
- (a) Mean QLIKE and MSE of each forecast
- (b) With true variance 1: QLIKE of forecasting 0.5 vs 2.0

Solution

- A: QLIKE 0.0231, 0.0945, 0.1108, mean 0.0762; MSE 0.150
- B: QLIKE 0.0457, 0.0020, 0.0945, mean 0.0474; MSE 0.030
- B wins on both: it follows the level of volatility
- QLIKE: $F = 0.5$ gives 0.307, $F = 2.0$ gives 0.193: under-prediction is punished more (MSE: 0.25 vs 1, the opposite)

A7: A Two-Day log-HAR Forecast [Proposed]

Solution discussed in class

Task

- ▣ Model of A5; RV over the last five days (today first): 2.0, 1.6, 1.3, 1.1, 1.0; monthly mean of logs $\ln 0.9$
- ▣ (a) Forecast for tomorrow
- ▣ (b) Forecast for the day after, iterating: treat tomorrow's log forecast as observed, drop the oldest day from the week, and update the monthly mean as $(21 \ln 0.9 + \hat{f}_1)/22$
- ▣ (c) Why does the forecast fall towards the long-run level?
- ▣ Model: A5

A8: Diebold–Mariano and Mincer–Zarnowitz Step by Step [Proposed]

Solution discussed in class

Task

- ▣ Lecture 9, SPY 2022–2026: mean QLIKE difference log-HAR minus GARCH $\bar{d} = -0.0327$, Newey–West standard error 0.0142
- ▣ (a) DM statistic and two-sided p -value
- ▣ (b) MZ for GARCH: $\hat{a} = 0.24$ (SE 0.23), $\hat{b} = 0.70$ (SE 0.24): test $a = 0$ and $b = 1$ separately
- ▣ (c) Is “no significant bias” the same as “a good forecast”?
- ▣ Model: A6

B1: Jumps in SPY [Solved]

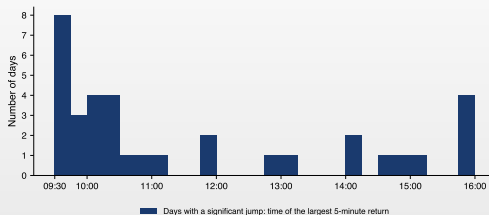
Task

- SPY, 5-minute returns, 1 489 days [Huang & Tauchen \(2005\)](#)
- (a) The ratio statistic z_t for every day; how many days reject at 5%, 1%, 0.1%?
- (b) Compare with the number expected under no jumps
- (c) On the 0.1% jump days: at what time is the largest 5-minute return?

Interpretation

- Are there more rejections than the level alone would give?
- Do the jump times point to scheduled news?
- Which level would you use to build a jump component for forecasting?

B1: Results [Solved]



- Rejections: 235 at 5% (expected 74.5), 110 at 1% (expected 14.9), 35 at 0.1% (expected 1.5); mean $z = 0.50$, s.d. 1.14
- Top days: 2021-08-27 ($z = 5.22$, jump 49% of RV, interval ending 10:05), 2022-01-03 ($z = 5.21$, ending 09:50), 2025-08-25 ($z = 4.77$, ending 16:00)
- Far more rejections than expected: real jumps, but also the test over-rejects when the intraday U-shape makes returns unequal
- Use 0.1%: few false jumps; the jump component is then 2.8% of total RV

B2: Jumps in Bitcoin [Proposed]

Solution discussed in class

Task

- ▣ Bitcoin, 5-minute returns, 659 full UTC days
- ▣ (a) Share of jump days at 0.1%; binomial test against the level
- ▣ (b) Weekends vs weekdays: share of jump days and mean BV/RV
- ▣ (c) Jump share of total variation
- ▣ Model: B1

Interpretation

- ▣ Why are jump days more frequent than for SPY, while jumps carry less variance?
- ▣ What makes a calm weekend day “jumpy”?
- ▣ Would you model weekend and weekday volatility together?

B3: Which Sampling Frequency? [Solved]

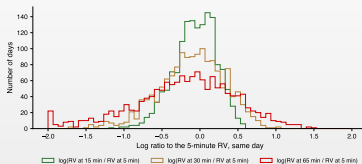
Task

- SPY: RV from returns every 5, 15, 30 and 65 minutes (one grid and the average over shifted grids)
- (a) Mean annualised volatility at each frequency
- (b) Ratio of mean RV to the 5-minute RV with a moving-block bootstrap 95% CI (blocks of 20 days) [Künsch \(1989\)](#)
- (c) Correlation of today's log RV with tomorrow's 5-minute log RV

Interpretation

- Is there a noise bias at 5 minutes?
- Why a block bootstrap and not an i.i.d. one?
- Which frequency gives the most useful measurement for forecasting?

B3: Results [Solved]



Interval	Vol. (one grid)	Vol. (avg. grids)	Ratio [95% CI]	Corr. next day	idem, avg.
5 min	12.93	12.93	1.000 [1.000, 1.000]	0.73	0.73
15 min	12.88	12.79	0.992 [0.954, 1.035]	0.71	0.72
30 min	12.75	12.77	0.973 [0.933, 1.015]	0.70	0.72
65 min	12.76	12.88	0.974 [0.913, 1.037]	0.64	0.70

- All CIs contain 1: no measurable noise bias for SPY at 5 minutes (consistent with the flat signature plot)
- Block bootstrap: daily RVs are strongly autocorrelated; i.i.d. resampling would make the CI too narrow
- Predictive correlation falls from 0.73 (5 min) to 0.64 (65 min, one grid): sparse RV is noisier; averaging shifted grids recovers part of it (0.70)

B4: Returns Standardised by Realised Volatility [Proposed]

Solution discussed in class

Task

- SPY open-to-close and Bitcoin daily returns Andersen et al. (2001)
- (a) Kurtosis of $r_t/\text{sd}(r)$ and of $r_t/\sqrt{\text{RV}_t}$, each with a block bootstrap 95% CI
- (b) Jarque–Bera test and the share of $|z| > 3$ compared with the Normal 0.27%
- Model: B3 (block bootstrap)

Interpretation

- Where do the heavy tails of daily returns come from?
- Why is the kurtosis of $r_t/\sqrt{\text{RV}_t}$ slightly **below** 3?
- What does this imply for a VaR model (Chapter 7)?

B5: log-HAR for SPY with Robust Inference [Solved]

Task

- ▣ SPY total daily variance, 1 467 days after the first month [Corsi \(2009\)](#)
- ▣ (a) OLS log-HAR; usual and Newey–West standard errors (7 lags)
- ▣ (b) Wald test $\beta_w = \beta_m$
- ▣ (c) Ljung–Box test on residuals (22 lags) and on log RV itself; compare with an AR(1)

Interpretation

- ▣ Which component matters most?
- ▣ Do robust standard errors change any conclusion?
- ▣ Has HAR captured the long memory?

B5: Results [Solved]

	$\hat{\beta}$	SE OLS	SE NW	t_{NW}
β_0	-0.079	0.025	0.027	-3.0
β_d	0.362	0.031	0.037	9.8
β_w	0.337	0.049	0.048	7.0
β_m	0.167	0.046	0.044	3.8

- ▣ $R^2 = 0.470$ vs 0.413 for an $AR(1)$ ($\hat{\phi} = 0.643$); persistence 0.866
- ▣ Daily component largest, but weekly and monthly are strongly significant; NW errors are up to 20% larger, no conclusion changes
- ▣ Wald $\beta_w = \beta_m$: 4.17 ($p = 0.041$): the weekly weight is larger
- ▣ Ljung-Box(22): log RV 4641, residuals 17.5 ($p = 0.73$): three coefficients absorb the long memory

B6: Forecasting the Next Week [Proposed]

Solution discussed in class

Task

- SPY, target $\sum_{j=0}^4 \text{RV}_{t+j}$ (next 5 days), forecasts made the evening before t , from January 2022
- (a) Direct log-HAR regression of the 5-day sum; expanding window
- (b) GARCH(1,1)- t : $\sum_{h=0}^4 [\bar{\sigma}^2 + (\alpha + \beta)^h (\sigma_t^2 - \bar{\sigma}^2)]$
- (c) QLIKE, MSE, DM test, MZ regression
- Model: B5, A6, A8

Interpretation

- Does the advantage of log-HAR survive at a weekly horizon?
- Overlapping 5-day targets: what does this do to the DM test?
- Which model is biased, and in which direction?

B7: Do Jumps Help? The HAR-CJ Model [Proposed]

Solution discussed in class

Task

- SPY; continuous part $C_t = \text{RV}$ without significant jumps (0.1%) plus squared overnight return; jump part J_t Andersen, Bollerslev & Diebold (2007)
- (a) Regress $\ln$ total variance on $\ln C$ (day, week, month) and $\ln(1 + J_t)$, with Newey–West errors
- (b) Out-of-sample QLIKE vs log-HAR from 2022; DM test
- Model: B5, B1

Interpretation

- Is the jump coefficient significant? What sign did you expect?
- Why can a significant coefficient give no forecasting gain?

B8: How Rough Is Volatility? [Proposed]

Solution discussed in class

Task

- SPY; $\ln \sigma_t = \frac{1}{2} \ln(\text{measure})$ for four measures: total variance, intraday RV, BV, 30-minute subsampled RV
Gatheral, Jaisson & Rosenbaum (2018)
- (a) $\hat{H}$ from $m(q, \Delta)$, $q \in \{0.5, 1, 1.5, 2, 3\}$, $\Delta = 1, \dots, 30$
- (b) Block bootstrap 95% CI (blocks of 60 days, 300 resamples)
- (c) Simulate a Brownian motion ($H = 0.5$), add noise, and estimate H again
- Model: B3 (block bootstrap)

Interpretation

- Is H below 0.5 for every measure?
- What does measurement error do to $\hat{H}$?

C1: Is Bitcoin's Volatility More Predictable Than the S&P 500's? [Proposed]

- Same out-of-sample window for both: March 2025 – September 2026
 - ▶ log-HAR, GARCH(1,1)- t and yesterday's RV for SPY (total variance) and Bitcoin (24-hour RV)
- Define “predictable”: R^2 of log RV on the log forecast, QLIKE gain over yesterday's RV, gain over GARCH
- Put a block bootstrap CI on the difference in R^2 between the two assets
- Improve the Bitcoin model with what you learned in B2 (weekends); does the answer change?
- Project extension: add Ethereum or a BVB stock, and the [Pele & Mazurencu-Marinescu-Pele \(2019\)](#) entropy measure as a regressor

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "I have SPY trade prices, every tick from 09:30 to 16:00, in a pandas Series px with a time index. Compute the daily realised variance, the annualised volatility and the HAR regressors."

The AI answer: text

- $RV_t = \sum_i r_{t,i}^2$, with $r_{t,i}$ the intraday log returns of day t [Andersen et al. \(2001\)](#)
- Use every tick: more returns always make RV_t more precise
- RV_t is SPY's daily variance: compare it directly with squared close-to-close returns and with the GARCH variance
- HAR: regress RV_{t+1} on RV_t and on its 5-day and 22-day averages [Corsi \(2009\)](#)

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np, pandas as pd
lp = np.log(px)
r = lp.groupby(lp.index.date).diff().dropna()
RV = (r**2).groupby(r.index.date).sum()
vol_ann = np.sqrt(RV) * 252
H = pd.DataFrame({'d': RV,
                  'w': RV.rolling(5).mean(),
                  'm': RV.rolling(22).mean(),
                  'y': RV.shift(-1)}).dropna()
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ The answer contains three errors; everything else is correct: find them
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ What does A3 say about $\mathbb{E}[RV^{(n)}]$ when prices contain noise?
- ▣ What happens to SPY between 16:00 and 09:30 the next morning?
- ▣ Plug in a typical day, $RV_t = 1.0 \times 10^{-4}$ (1%²): is the annualised volatility plausible for SPY?

Common Pitfalls

- ▣ Sampling every tick without checking the signature plot
- ▣ Forgetting the overnight return: intraday RV understates the daily variance of SPY by about a third
- ▣ Treating RV as the true variance: it has a standard error, and it makes R^2 of forecasts look low
- ▣ Comparing volatility forecasts with MAE or with MSE on volatility: not robust to the noisy proxy
- ▣ Usual OLS standard errors in HAR and i.i.d. bootstrap for daily RV: both ignore autocorrelation
- ▣ Reading every significant jump test as a news jump: thousands of tests give false alarms

Take-Home Assignment (Team Project)

- Choose an asset with intraday data not covered in class (Ethereum, a currency pair, a stock)
- Build daily RV, BV and a jump component; show the signature plot and the intraday pattern
- Compare log-HAR (with at least one extension) and GARCH out of sample with QLIKE, DM and MZ
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Nonparametric and Semiparametric Models
- ▣  Applied Time Series Analysis with Python

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_09

References

- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X. (2007). [Roughing it up: including jump components in the measurement, modeling, and forecasting of return volatility](#). *Review of Economics and Statistics*, 89(4), 701–720.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2001). [The distribution of realized exchange rate volatility](#). *Journal of the American Statistical Association*, 96(453), 42–55.
- ▣ Bandi, F.M., Russell, J.R. (2008). [Microstructure noise, realized variance, and optimal sampling](#). *Review of Economic Studies*, 75(2), 339–369.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2004). [Power and bipower variation with stochastic volatility and jumps](#). *Journal of Financial Econometrics*, 2(1), 1–37.
- ▣ Corsi, F. (2009). [A simple approximate long-memory model of realized volatility](#). *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy](#). *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). [Volatility is rough](#). *Quantitative Finance*, 18(6), 933–949.
- ▣ Huang, X., Tauchen, G. (2005). [The relative contribution of jumps to total price variance](#). *Journal of Financial Econometrics*, 3(4), 456–499.
- ▣ Künsch, H.R. (1989). [The jackknife and the bootstrap for general stationary observations](#). *Annals of Statistics*, 17(3), 1217–1241.
- ▣ Patton, A.J. (2011). [Volatility forecast comparison using imperfect volatility proxies](#). *Journal of Econometrics*, 160(1), 246–256.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). [Using high-frequency entropy to forecast Bitcoin's daily value at risk](#). *Entropy*, 21(2), 102.
- ▣ Zhang, L., Mykland, P.A., Aït-Sahalia, Y. (2005). [A tale of two time scales: determining integrated volatility with noisy high-frequency data](#). *Journal of the American Statistical Association*, 100(472), 1394–1411.